

Labreport Advanced Computer Architecture
Lab 4

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Class: MEIsw

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1 Introduction

In this lab the different memory structures and access methods on the GPU will be explored. First we will look at memory coalescence, afterwards we will look at the difference between global, constant and shared memory. The source code for this project can be found on `https://github.com/RoelsJonas/Advanced-Computer-Archticture-Lab-4`

2 Coalesced memory

For this test case a program was developed that converted a color image to a gray scale image by taking the average of the red, green and blue values of every pixel. To use coalescing in the program the image was first converted to first contain all red subpixels, then the green ones and finally the blue subpixels. This way subsequent threads will read subsequent values thus making use of memory coalescence. The results of this experiment can be found in Figure 1. Here we see that using coalescence is almost 3 times faster on average.

During this experiment we noticed that the first couple of runs of the kernel were always slower. This is also demonstrated in Figure 2. Here we see that after approximately 25 runs the processing time has stabilized. Therefor for all experiments in this report we first did 50 warm up runs of which we discarded the results.

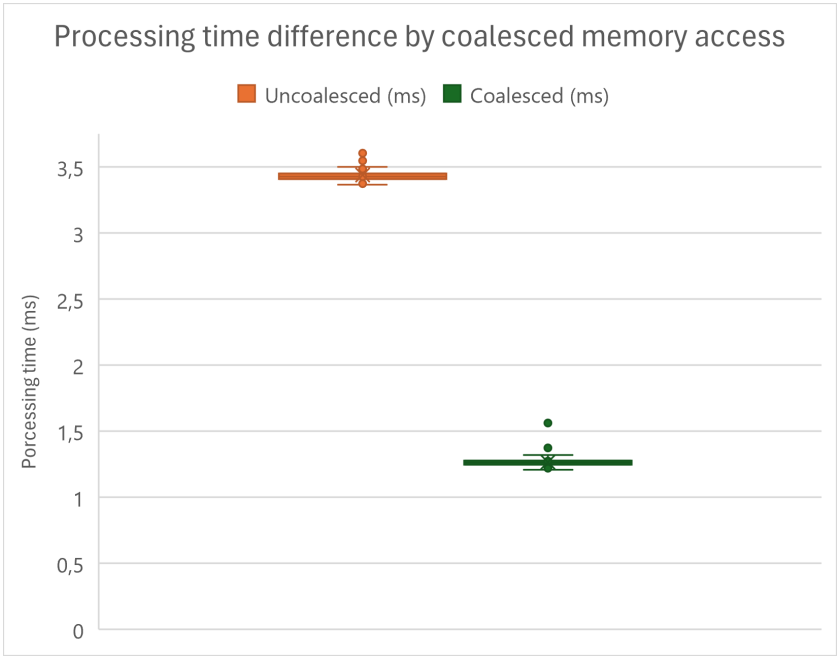


Figure 1: Comparison of the processing time between coalesced and uncoalesced memory access

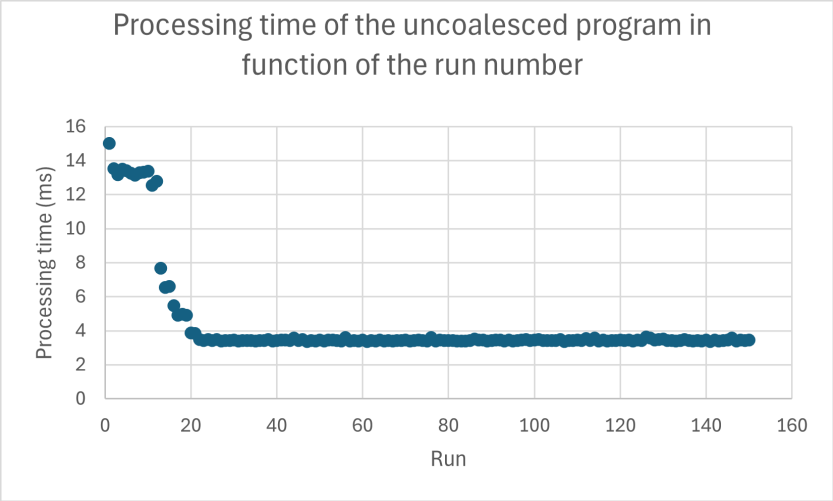


Figure 2: Processing time of the uncoalesced program during different runs

3 GPU memory types

In the second part of the lab assignment we compared the use of different memory types when multiplying matrices.

Looking at figure 3 down below we multiplied two matrices of size 64 by 64. As expected the shared memory results in the fastest average processing time. The average processing time when using constant memory is just a bit slower and closely behind is the global memory approach.

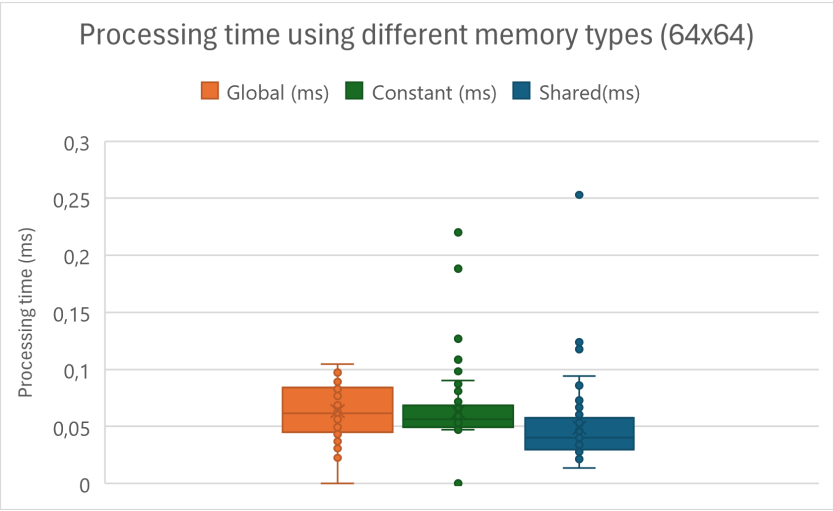


Figure 3: Comparison of the processing time using the different gpu memory types for a 64x64 matrix multiplication

Next we increased 4 the matrix to the maximum size of 88 by 88. This is the maximum because the constant memory has a size of 64kB which can store 2 matrices of 32kB and thus resulting in a matrix with 88 rows and columns. [1] The results are the same as before, namely that shared memory is again a bit faster than both global and constant memory. However now the global approach is slightly faster then the constant approach.

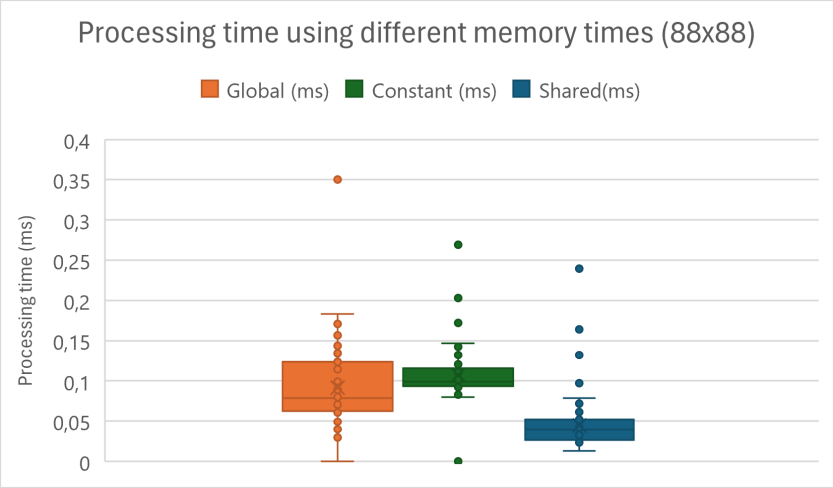


Figure 4: Comparison of the processing time using the different gpu memory types for a 88x88 matrix multiplication

Finally, in Figure 5 we see the results of another comparison between the shared and global memory. Here we increased the matrix sizes to 1024 by 1024 as we weren't limited by the capacity of the constant memory. Here we can really see the benefits of shared memory over global memory.

[1] While the size of the shared memory is only 40kB, this doesn't form a problem as only certain sections (required for the calculations done by the current block) are copied to it.

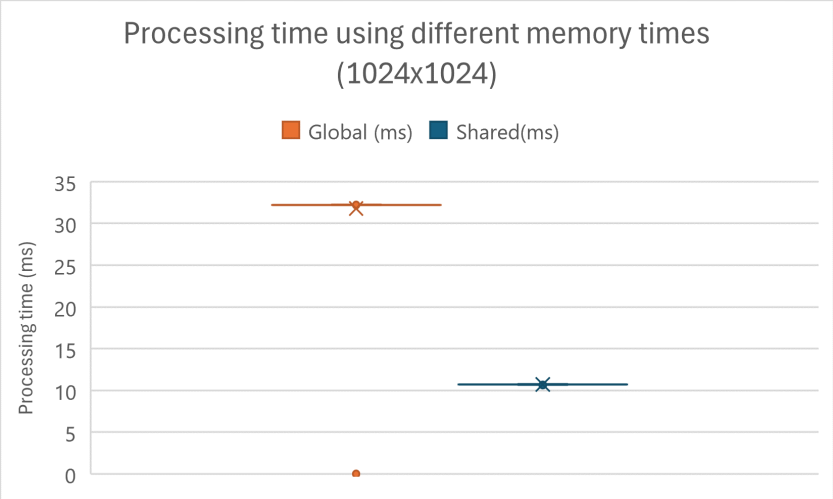


Figure 5: Comparison of the processing time using the different gpu memory types for a 1024x1024 matrix multiplication